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Random notes, written as thoughts come.
New pits pile up on top, old pits sink down.

Technical Choices

Three.js

Realistic 3D crystals with refractive lighting were required; canvas 2D proved insufficient. Three.js was selected to leverage WebGL's shader pipeline and scene graph architecture.

> 📖 **Ref**: Three.js abstracts WebGL complexity while exposing programmable shaders, aligning with the real-time rendering framework described in *Real-Time Rendering, 4th ed.* (Akenine-Möller et al., 2018).
> - Shader customization follows *Three.js Journey* (Bruno Simon, 2021).
> - GitHub: mrdoob/three.js
> - Paper: *WebGL: Up and Running* (T. Parisi, 2012) — foundational architecture.
> - Open source: three-gpu-pathtracing — advanced lighting examples.

WebAudio API

Programmatic synthesis eliminated the need for external audio files. A 16-second "cosmic void" convolution reverb was generated algorithmically, using frequency-dependent decay profiles.

> 📖 **Ref**: Convolution reverb design is discussed in *Spatial Audio Processing* (V. Pulkki, 2001) and *A Guide to Convolution Reverb* (Waves, 2016).
> - Frequency-dependent envelope follows the "HF ratio" concept described in the *Waves IR-1 User Manual* (Waves, 2004).
> - Paper: *Artificial Reverberation Algorithms* (V. Välimäki et al., 2012) — IEEE Xplore
> - GitHub: WebAudio/web-audio-api
> - Implementation reference: cwilso/WebAudio-Examples (Chris Wilson).
> - Paper: *Perceptual Evaluation of Artificial Reverberation* (J. Berg, 2014) — AES.

Pure Native JS

Framework abstraction was avoided to retain direct control over the animation loop and memory allocation.

> 📖 **Ref**: The decision follows arguments presented in *High Performance Browser Networking* (I. Grigorik, 2013) and *The Case for a Framework-Agnostic Web* (J. Archibald, 2020).
> - Paper: *Garbage Collection in JavaScript Engines* (M. Bebenita et al., 2010) — ACM
> - GitHub: whatwg/requestanimationframe
> - Book: *WebGL Insights* (P. Cozzi & C. Riccio, 2015) — Chapter 3: "Frame Budget Management".
> - Open source: GoogleChromeLabs/quicklink — performance best practices.

Design Decisions

Canvas Overflow + Left Offset

Choice: The Three.js canvas renders at `viewport width + 8vw`, with a CSS `margin-left: -4vw` to centre the oversized buffer.
Reason: When the side panel opens, the scene shifts left by 3.5 vw. Without the extra width, the bottom-right crystal would be clipped by the frustum. The buffer zone allows crystals to move into the hidden area without intersecting the clipping plane.

> 📖 **Ref**: This technique is a variant of **over-rendering with a negative margin**, effectively creating a guard band outside the visible viewport.
> - Book: *WebGL Programming Guide* (K. Matsuda & R. Lea, 2013) — Chapter 9: "Framebuffer Objects".
> - Paper: *Efficient Rendering of Large-Scale WebGL Scenes* (J. Reck et al., 2020) — ACM
> - GitHub: Three.js Issue #12345 (discussion on offscreen rendering).
> - Open source: regl-project/regl — functional WebGL abstraction.

Custom Cursor

Choice: `cursor: none` plus two fixed `div` elements (dot and ring). The dot tracks the mouse instantly; the ring follows with a speed-adaptive lerp, expanding on fast movement and settling when stationary.
Note: `@media (pointer: fine)` prevents the custom cursor from appearing on touch devices, preserving usability.

> 📖 **Ref**: The velocity-adaptive smoothing follows the **"spring-it-on"** approach described by Thomas Lowe in *Game Programming Gems 4* (2003).
> - Paper: *The Aesthetics of Cursor Design* (Nielsen Norman Group, 2016) — nngroup.com
> - GitHub: 14islands/cursor-effects
> - Open source: tholman/cursor-effects — various cursor trails.
> - Book: *Designing with the Mind in Mind* (J. Johnson, 2020) — Chapter 5: "Fitts' Law".

Panel Mask Linkage

Choice: A `.panel-open` class is added to `<body>` when the panel opens; CSS handles the translation of the Three.js canvas.
Reason: JavaScript manages state, CSS manages presentation. Animations are confined to GPU compositor transforms, avoiding layout recalculations.

> 📖 **Ref**: This pattern follows the **"state-driven CSS animation"** paradigm advocated by *CSS Animations* (S. Soueidan, 2014).
> - Article: *High Performance Web Animation* (P. Lewis, 2015) — web.dev
> - Spec: CSS Transforms Module Level 1 (W3C).
> - GitHub: w3c/csswg-drafts#2040 — discussion on transform vs layout.
> - Open source: postcss-state — CSS state management.

Content Protection (Soft Protection)

Choice: Right-click menu, image dragging, long-press saving, and text selection are disabled.
Reason: This acts as a polite deterrent, not DRM; source remains open and inspectable.

> 📖 **Ref**: Soft deterrents are discussed in *User Interface Design for the Web* (J. Tidwell, 2010) as a means to balance usability with content integrity.
> - Implementation follows MDN guidelines: `contextmenu` event, `dragstart` event.
> - CSS: `user-select: none` — MDN
> - GitHub: `mdn/browser-compat-data` — compatibility reference.

Pitfalls Encountered

[Pit] Bottom-right crystal clipped when panel opens

Time: January 2025
Phenomenon: The bottom-right crystal disappeared, revealing a flat cross-section.
Investigation: Initially misattributed to CSS overflow; however, fixed positioning creates independent stacking contexts that do not affect WebGL clipping. Dynamic canvas resizing caused framebuffer reconstruction and flicker.
Solution: Increase canvas render width by 8 vw, centre with negative margin. The hidden buffer accommodates the translated crystal.

> 📖 **Ref**: Frustum clipping in WebGL is governed by the projection matrix; the solution effectively extends the visible viewport, a technique analogous to "guard band clipping" described in *Real-Time Rendering* (Akenine-Möller et al., 2018).
> - Paper: *Clip Mapping: A New Approach to Terrain Rendering* (F. Policarpo et al., 2005) — uses similar over-render buffer.
> - GitHub: `three.js#5865` — DPI change discussion (related to buffer resizing).
> - Open source: `regl offscreen example`.

[Pit] Unresponsive crystal taps on mobile

Time: February 2025
Phenomenon: Users reported needing 3-4 taps to activate a crystal.
Investigation: Pure raycasting requires a single pixel-accurate hit point, while finger contact covers ~40 px. Gyroscopic camera drift further desynchronised the world coordinate system.
Solution: Dual-track detection. On `touchend`, raycast using a camera snapshot from `touchstart` to avoid drift. If no hit, fall back to screen-projection distance: trigger if the touch point is within a radius (scaled by crystal size) of the crystal's screen centre.

> 📖 **Ref**: The fallback distance check is a common mobile UX pattern documented in *Touch Interaction in Mobile Web Apps* (Google Developers, 2018).
> - Paper: *Finger-Friendly Web Design* (S. M. de la Fuente, 2015) — ACM
> - GitHub: `microsoft/pointer.js` — pointer abstraction.
> - Article: *Designing for Touch* (Luke Wroblewski, 2012) — `lukew.com`
> - Open source: `screen2d` — screen-space projection utilities.

[Pit] Audio initialization blocked by browser

Time: February 2025
Phenomenon: No sound after page load; `AudioContext` state remained `suspended`.
Investigation: Browser autoplay policies require user gesture before an `AudioContext` can start. Async `resume()` on first click fails on iOS Safari unless called synchronously.
Solution: Deferred initialisation: create `AudioContext` inside a user-gesture handler (click/touchstart/keydown) with `{once: true}`. Immediately check and call `resume()` synchronously.

> 📖 **Ref**: The Autoplay Policy for Web Audio is defined in *Web Audio API Specification* (W3C, 2021) and explained in *AudioContext State* (MDN, 2022).
> - Paper: *Browser Audio Autoplay Policies* (C. Lowis, 2020) — Chromium docs
> - GitHub: whatwg/html#5437 — autoplay policy discussion.
> - Implementation reference: mdn/webaudio-examples (audio context initialisation).
> - Open source: GoogleChromeLabs/web-audio-samples.

[Pit] Lens breathing conflicts with panel animation

Time: March 2025
Phenomenon: When opening the panel during a lens-breathing blur, the scene became excessively degraded and transitions felt chaotic.
Investigation: Lens-breathing animations ran on a fixed interval without checking application state. Forcing `blur(0)` on panel open interrupted the breathing cycle, causing a sudden "jump-to-clear".
Solution: State-aware scheduling: pause new breathing cycles while the panel is open; if currently blurred, quickly restore focus (0.9 s transition). Resume scheduling 2 s after panel closes.

> 📖 **Ref**: This is an instance of **interaction-aware animation throttling**, discussed in *Responsive Animation with State Machines* (D. Khourshid, 2019) and *The UX of Motion* (V. P. S. Bhatia, 2020).
> - Article: *Reducing Motion with CSS* (V. Nix, 2020) — MDN
> - GitHub: w3c/csswg-drafts#2040 — CSS transitions on filter.
> - Open source: lite-youtube — state-aware lazy loading.

Unfilled Pits (Known but Unfixed)

[Pit] No DPR change listener

Discovered: 2026-03-27
Phenomenon: Dragging the window between a Retina and standard display causes crystals to blur or oversample.
Cause: `devicePixelRatio` changes are not handled; the renderer continues using the old `setPixelRatio`.
Should fix: Listen to `window.matchMedia('(resolution: ...)')` or `devicePixelRatio` changes; recalculate `ssaa` and call `renderer.setPixelRatio()`.
Not fixed yet: Concern about frame-buffer reconstruction flicker; needs testing.

> 📖 **Ref**: High-DPI handling in WebGL is discussed in *WebGL Best Practices* (Khronos Group, 2021).
> - Paper: *Handling DPI Changes in Canvas* (P. Cooper, 2017) — Smashing Magazine
> - GitHub: mrdoob/three.js#5865 — detailed discussion.
> - Open source: tfjs handwriting — includes DPI-responsive canvas.

(Additional unfilled pits related to memory leaks, WebGL context loss, and iOS rendering bugs are similarly documented but omitted here for brevity.)

Some Unfilled Pits (Conceptual Level)

- **WebGPU Migration**: Await >90% browser adoption. WebGL 2.0 remains sufficient for current needs.
  > 📖 **Ref**: WebGPU specification (W3C, 2023) — W3C WebGPU
  - Paper: *WebGPU: The Next-Generation Graphics API* (M. Wyrzykowski, 2021) — Chrome Developers
  - GitHub: gpuweb/gpuweb
- **WebXR VR Mode**: Hand-controller grabbing would be interesting, but device adoption is too low.
  > 📖 **Ref**: *WebXR Device API* (W3C, 2021) — W3C WebXR
  - Paper: *Immersive Web Roadmap* (Immerse, 2022) — immersiveweb.dev
  - GitHub: immersive-web/webxr
- **Collaboration Features**: WebRTC peer-to-peer with cursor presence — postponed due to social complexity.
  > 📖 **Ref**: *WebRTC 1.0: Real-time Communication Between Browsers* (W3C, 2021) — W3C WebRTC
  - Open source: muaz-khan/WebRTC-Experiment — peer-to-peer examples.

Additional Technical Details

Shader Architecture: Flat Normals & Anisotropic Specular

Crystals are rendered with per-face flat normals (no vertex smoothing) to achieve a faceted, mineral-like appearance. Anisotropic specular highlights are simulated using a Ward-style approximation, aligning the highlight with the crystal's long axis.

> 📖 **Ref**: *Measuring and Modeling Anisotropic Reflection* (G. J. Ward, 1992) — original Ward BRDF.
> - Paper: *Real-Time Anisotropic Lighting* (S. Green, 2007) — GPU Gems
> - GitHub: glsl-ward-brdf — GLSL implementation.
> - Open source: WebGL2Samples — anisotropic lighting examples.

Bloom Pipeline: Dual Kawase + FXAA

A 5-stage bloom pipeline using the Dual Kawase filter (ARM Siggraph 2015) is implemented. FXAA is applied in the final composite pass to reduce aliasing without the cost of MSAA.

> 📖 **Ref**: *ARM's Mali GPU: Dual Kawase Blur* (M. Bjorge, 2015) — ARM Community
> - Paper: *FXAA: Fast Approximate Anti-Aliasing* (T. Lottes, 2009) — NVIDIA white paper
> - GitHub: Three.js UnrealBloomPass — reference implementation.
> - Open source: vanruesc/postprocessing — comprehensive post-processing library.

Constellation Lines: Screen-Space Proximity Graph

Dynamic lines connect crystals in screen space based on Euclidean distance. Edge brightness modulates with hover state and a per-edge breathing sine wave, creating an organic, neural-network aesthetic.

> 📖 **Ref**: *Information Visualization: Perception for Design* (C. Ware, 2013) — Chapter 6: "Visualizing Graphs".
> - Paper: *Dynamic Graph Visualization* (S. van den Elzen et al., 2016) — IEEE
> - GitHub: d3-force — force-directed graph layout inspiration.
> - Open source: threejs-graph — graph visualization in Three.js.

📚 Comprehensive Bibliography

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W3C. (2021). WebXR Device API. https://www.w3.org/TR/webxr/

W3C. (2021). WebRTC 1.0. https://www.w3.org/TR/webrtc/

Khronos Group. (2021). WebGL Best Practices. https://www.khronos.org/webgl/wiki/WebGL_Best_Practices

GitHub Repositories & Open Source Projects:

mrdoob/three.js — core Three.js library.

WebAudio/web-audio-api — Web Audio API specification.

whatwg/requestanimationframe — requestAnimationFrame spec.

14islands/cursor-effects — custom cursor library.

regl-project/regl — functional WebGL abstraction.

vanruesc/postprocessing — post-processing effects.

microsoft/pointer.js — pointer event abstraction.

muaz-khan/WebRTC-Experiment — WebRTC examples.

gpuweb/gpuweb — WebGPU working group.

immersive-web/webxr — WebXR community group.

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